

$$\bar{X}_n + \text{Buffer}_n < 30.$$

So if this is still true, you would conclude that waiting time is less than 30 minutes.

6.8 Example:

→ 32% of Americans watch their news on youtube.

Can you conclude that at most a third of Americans watch their news on youtube?

we are using at-least/at-most interchangeably.

It's a $\text{ber}(p)$, I make my modelling assumptions here

To Generalise it, I would standardize it, make it a Gaussian and compute variance $\bar{X}_n$.

→ I can do all this, if I really know p .

→ Assume $p = 33\%$ is true.

→ Standardize.

→ Asymptotic Variance.

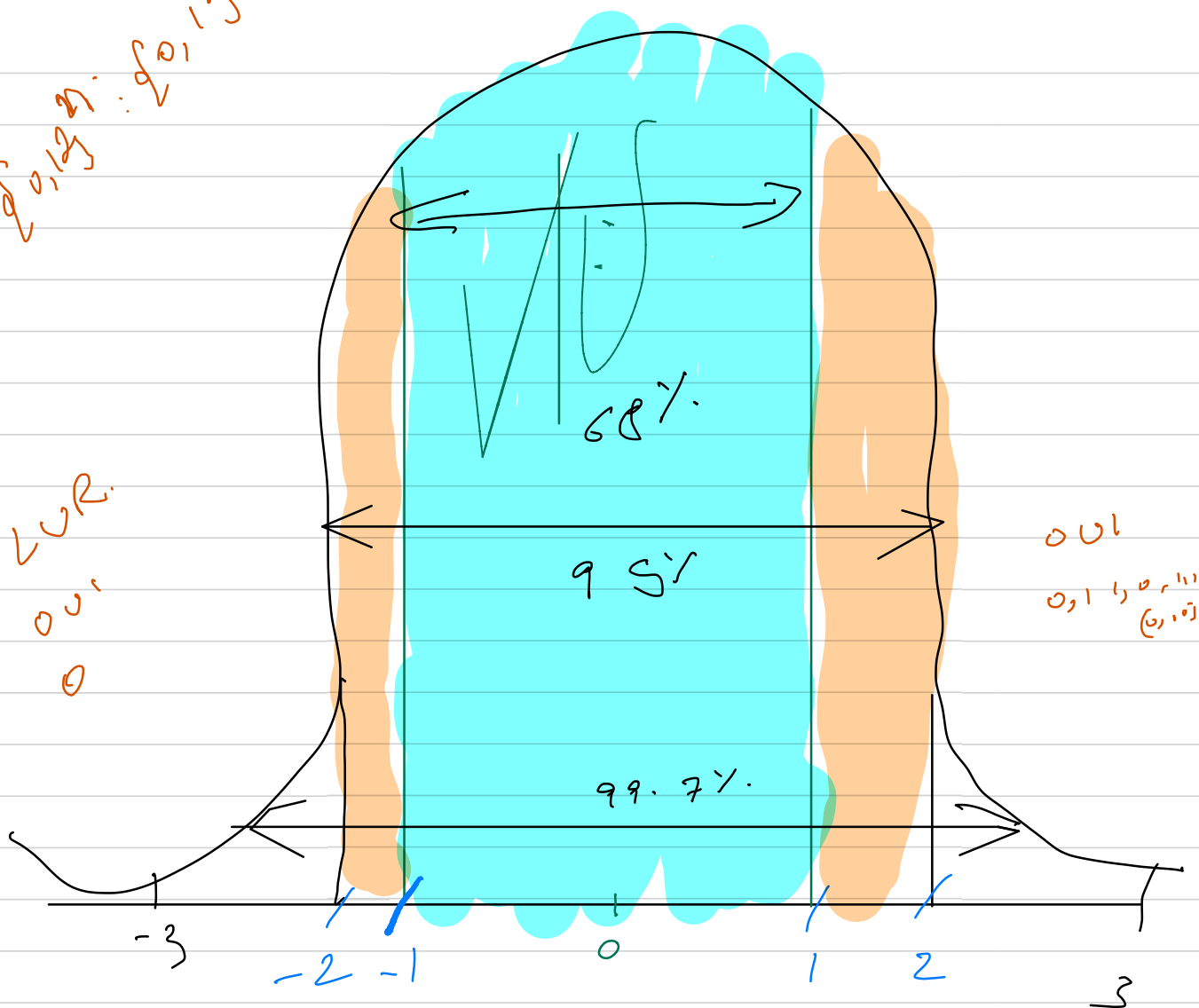
→ If I plug in $\bar{x}_n$, I see 5 million, it's not Gaussian.

So now I know,
my basic
assumption .33
is wrong.

90th percentile $\rightarrow 1.645$

going, going

LR.
0.01
0



if $\bar{X}_n = .33$ I would get 0.

By plugging in $\bar{X}_n = .32$, I got -1.5

which is okay, and can be explained
by the fluctuations of the Standard Gaussian.

∴ It's on the scale of proportion of people.
So, on the scale of 52%.

It might be on the scale of waiting times in terms of minutes. Boarding time.

Then taking those guys and mapping them on a scale which is in terms of % age of Gaussian.

SO, THE PERCENTAGE OF GAUSSIAN WILL BE SOME UNIVERSAL SCALE WE HAVE IN MIND.

→ we are willing to tolerate % of error,

CI → 99% → This is the scale at which the statistician should always be thinking,

Statistician has to ask the practitioner,
what % age of error are you willing to tolerate? → between (0, 1).

↓
map it back on the scale of problem.
(minutes / % age of Americans)

6.9 Fairness of a coin:

$\text{Ber}(p)$. 13 heads out of 30.

→ I want to test whether the coin is fair.

→ I could not make up as much data as I want.

i.i.d $\text{Ber}(p)$.

$$\bar{x}_n = .46.$$

→ If I saw 29 heads, it would not be a fair thing.

→ If I saw 15/16,

I should see something $N(0,1)$.

$$\sqrt{n} \frac{\bar{x}_n - .5}{\sqrt{.25}} \sim N(0,1) = .77.$$

↑
Plausible.

→ I started with the assumption of fair coin. $p = 1/2$.

I could not find contradiction.

I write, it is ^{not} unlikely the coin is fair.

6.10 STATISTICAL FORMULATION:

I know if my model is well specified and identifiable, I can go back and forth between a space of θ s & probability distributions.

$\rightarrow$ I take 2 subsets of $\theta \rightarrow \theta_0$ & θ_1 .

The question I am asking is does θ

Disjoint.

$(\theta_0 \cup \theta_1 \text{ is not } \Omega)$



$\theta_0 \rightarrow H_0 \rightarrow$ NULL HYPOTHESIS.

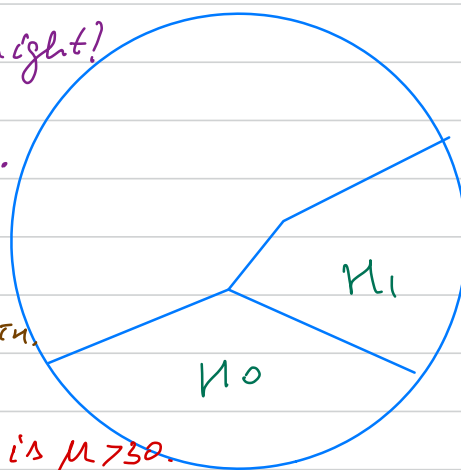
$\swarrow$
other one H_1 .

$\rightarrow$ Do people turn their head to the left or right?

$\rightarrow$ Is p different from $1/2$.

$\rightarrow$ Is waiting time < 30 at 11pm.

$\rightarrow$ we asked is $\mu > 30$.



→ Now, we want to ask, what does your parameter look like?

↳ But if my parameter is identifiable, I can go back and forth b/w probability distributions and parameters.

→ I am just asking about parameter

either in $(\theta_0 \text{ or } \theta_1)$ → Same sample space;
↓

$H_0 \rightarrow$ NULL HYPOTHESIS

$H_1 \rightarrow$ ALTERNATE HYPOTHESIS.

→ For PTH, $n > 30 \rightarrow H_1$

We are ^{not} testing H_0 against H_1 -
NOT SYMMETRIC.

→ We are looking for evidence against H_0 .

Hard: Am I a man above 35 or not?

Easy one: I am a man above 35 & am I a goat?

You are testing against H_0 but something very specific.
→ We want to decide whether we want to reject H_0 .

So, we are looking for evidence against H_0 , in favor of H_1 .

THE WAY TO THINK ABOUT IT:

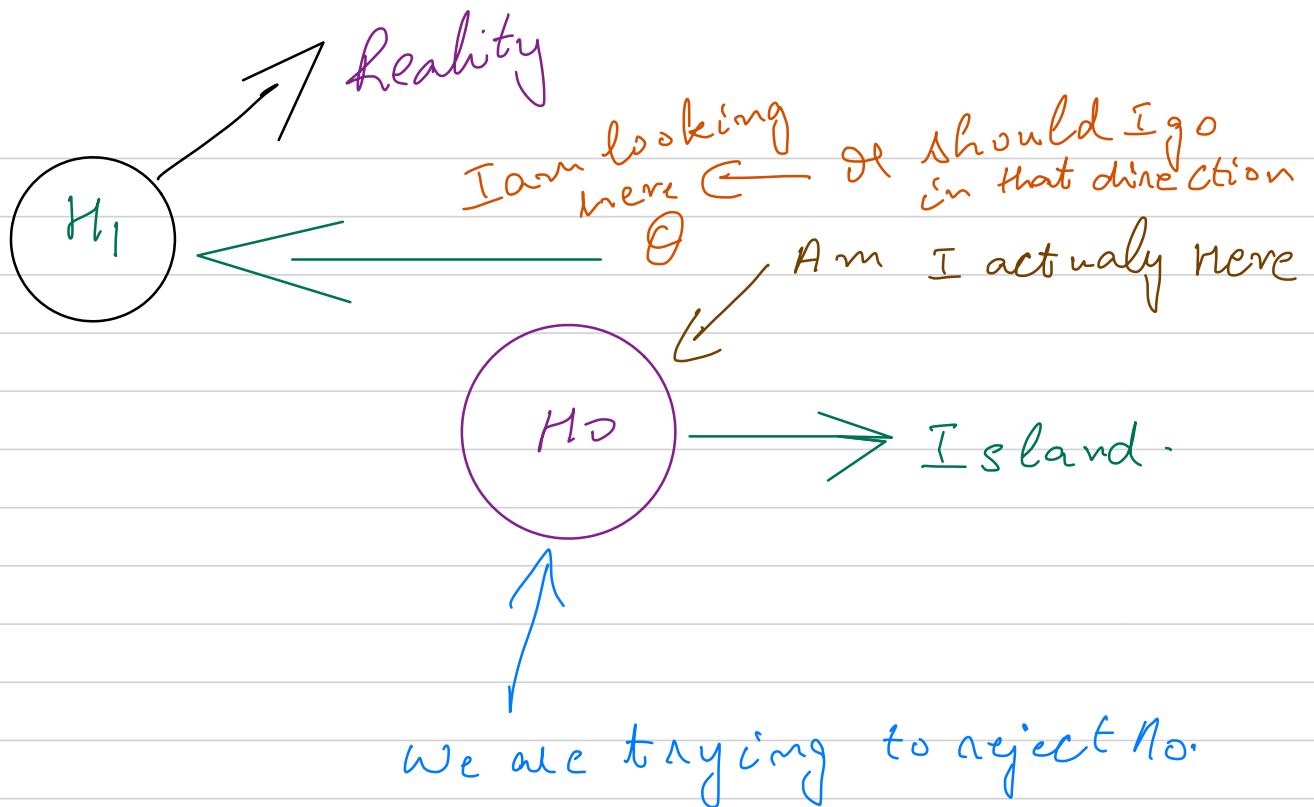
H_0 is a small island. - Go.

I am looking towards a particular direction which is H_1 . → This is my reality.

I want to say, is my theta telling me I actually here?

Is my θ telling me should I actually go in that direction?

YOU ARE TESTING AGAINST H_0
BUT SOMETHING VERY SPECIFIC.



We are never really trying to say H_0 is true.
 we are only saying, if we can find
 a contradiction.

So, let's assume H_0 is true, and we are trying
 to find a contradiction.

We have to very carefully decide which one
 is H_0 and which one is H_1 .

if we want to prove PPH has longer waiting times

↑ H_1 ↑ reality

I am trying to find evidence in the data, that I am worse than average.

By default, I should assume that PPH is worse than avg → H_1

INNOCENT UNTIL PROVEN GUILTY

Jury says not guilty, we did not find any evidence against you.

↳ If you walk away free, it's just we were not able to find enough evidence.

6.12 STATISTICAL TESTS:

→ I will have a binary answer at the end of the day, I might as well turn it into 0 or 1.

→ We are going to turn it into a fn. that maps it to either 0, if we want to conclude to H_0 or to 1, if we just want to conclude to H_1 .

It's ok to have a fn. that concludes at 0,
BUT THIS DOES NOT MEAN H_0 is true.
↳ It just means H_0 is not true.

A statistic is a fn. of the data and turn it into the number 0 or 1.



Test Statistic.

(Statistic can depend on the parameter).
↳ is measurable.

BUT A TEST, UNLIKE AN ESTIMATOR IS NOT LIVING IN THE SAME SPACE AS θ , IT IS JUST

0 or 1.

A test statistic is not living in the same space as $(\theta \text{ or } 0, 1)$, it is just 0 or 1.

↓ Intuition:

→ Any such fn. has to be an Indicator -

It's the indicator of where it's equal to 1.

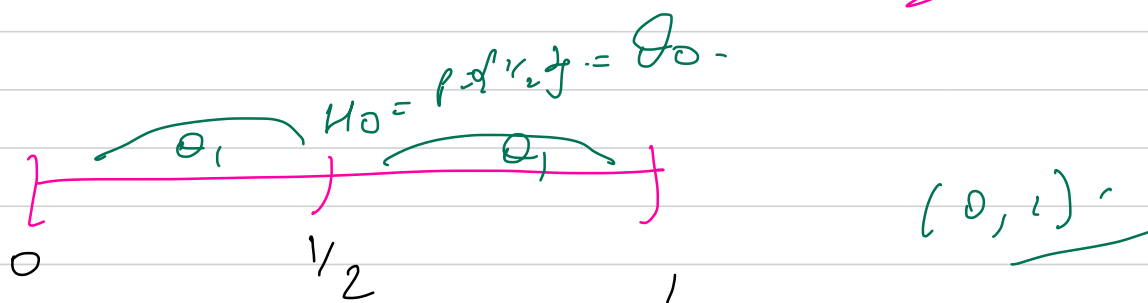
$$\psi(x) = \psi I(\psi(x) = 1) -$$

→ A test is a statistic so that $\psi \in \mathcal{G}_{0,1}$.

If $\psi = 0$, H_0 is not rejected.
 $\psi = 1$, H_0 is rejected.

↳ I am concluding to H_1 .

Coin Example: $H_0: p = 1/2$ vs $H_1: p \neq 1/2$



$$\varphi = \mathbb{I} \left\{ \sqrt{n} \frac{|\bar{X}_n - 1/2|}{\sqrt{1/2(1/2)}} > c \right\} \text{ for } c > 0$$

In the previous coin example

$$\frac{\sqrt{n} |\bar{X}_n - 0.5|}{\sqrt{0.5 \sqrt{0.5}}} \approx 0.77$$

I would get the absolute value.

I want to reject when this is large

I would have to decide whether it's above a threshold ~~or not~~.

O_0 and O_1 cover all of $[0, 1]$. They are disjoint. This is called a partition of $[0, 1]$.

TYPE I: ERROR.

→ INTUITION: Choosing the threshold C .

→ what am I trying to optimize?

✓ The performance should be measured by it's error.

→ oh, It can make an error.
✓ I believe H_0 is not true, but it's true.

✓ Those are the tests that always reject.

→ If I take a test that says, reject for $C=0$.

✓ This will always be larger than 0.
↳ will always reject.
↑ Pretty bad test.

→ So, we need to define, what kind of errors a test can make and try to minimize this.

I have a fn: It takes a set of n observations and maps it to 0 or 1.

Rejection Region is just the set of observations that would lead to the value 1.

ERRORS:

Rejection Region is a subset of sample space that would lead to rejection

Rejection Region of a test ψ :

$$R_\psi = \{x \in \mathcal{E}^m : \psi(x) = 1\} \Rightarrow \psi(x) = \mathbb{I}(x \in R_\psi)$$

$\mathcal{E}^m \rightarrow$ just where $(x_1, \dots, x_n)$ live.

Rejection Region is just a region that would lead to the value 1.

$$\psi(x) = \mathbb{I}(\psi(x) = \mathbb{I}(x \in R_\psi))$$

OR precisely the Indicator of the Rejection Region.

because the Region is exactly the set over which $\psi = 1$.

My test is only of 0, 1.

So either H_0 is true

or H_1 is true.

Rule:

Test Reality		H_0	H_1
H_0		✓	Type I
H_1		Type II ↓ β	✓

→ Denoted by α .

If the test says H_0 when it's H_0 , it's good.

And if the test says H_1 , when it's H_1 , it's good.

→ And one arises when the reality is H_0 , but I conclude to H_1 .

Remember
this:

Reality is H_0 but I reject it: Type I error.

Reality is H_1 , but I conclude to H_0 : Type II error.

↓
I think we say we reject H_0 .

$\alpha(\varphi)$: It is a fn. of the test.

↑ Function of making type I error.

$\alpha(\varphi): \theta_0 \rightarrow \mathbb{R} \cdot [0, 1)$.

$$\theta \rightarrow P_\theta[\varphi=1]$$

↑
probability of type I error.

So, you are quantifying what is this type of error.

→ The thing is H_0 is just one type of error.

In the coin example,

H_0 was just p is equal to $1/2$, which just one single reality.

$$\theta_0 \rightarrow \{1/2\}.$$

→

when $H_0: \mu \leq 50$

$H_1: \mu > 50$

$\mu \in \mathbb{R}.$

We have to quantify

$\theta \rightarrow P_\theta(\varphi=1)$ = quantify the value of type I error

Question: Compute the Type I error.

Let $x_i \stackrel{iid}{\sim} N(\mu, 1)$, where μ is an unknown parameter.

Question of Interest: Does $\mu = 0$?

Null Hypothesis: $H_0: \mu = 0$.

Alternate $H_1: \mu \neq 0$.

$$\psi_C = 1(\sqrt{n} |\bar{X}_n| > C).$$

Type I error, also known as Type I error rate of a test ψ is the fn.:

$$\alpha_\psi: \theta_0 \rightarrow [0, 1].$$

$$\theta \rightarrow P_\theta(\psi = 1).$$

If you choose the threshold, $C = 1.96$, what is the type I error?

(In this case, since H_0 consists of only one point, the fn. α_ψ is defined only at one point

and we use the terminology "type 1 error" to mean the value α_p at that point.

Soeⁿ:

If we assume the null hypothesis

$H_0: \mu=0$, and $\sigma^2=1$, it gives

$\sqrt{n} \bar{x}_n \sim N(0,1)$ for large n .

The probability of a type 1 error is

$$\alpha_p(0) = P_0(\varphi_C=1) = P(\sqrt{n}|\bar{x}_n| > q_{0.5}) = 0.1.$$

If H_0 is true, i.e. $\mu=0$, then $\sqrt{n} \bar{x}_n$ is

asymptotically normal.

$$|Z| = 0.05.$$

$$|Z| = 0.05 = 0.1$$

$$Z = 0.05$$

$$\therefore P_0(\varphi_C=1) = P_0(\sqrt{n}|\bar{x}_n| > q_{0.05})$$

the probability that H_0 is rejected even though it is true.

6.15 Type 2 Error and Power of a Statistical Test.

→ It says when H_1 is true, there might be several possible values of H_1 .

Type 2 error:

Not rejecting H_0 although H_1 is true.

$$\beta_\psi : \Theta_1 \rightarrow \mathbb{R} [0,1]$$

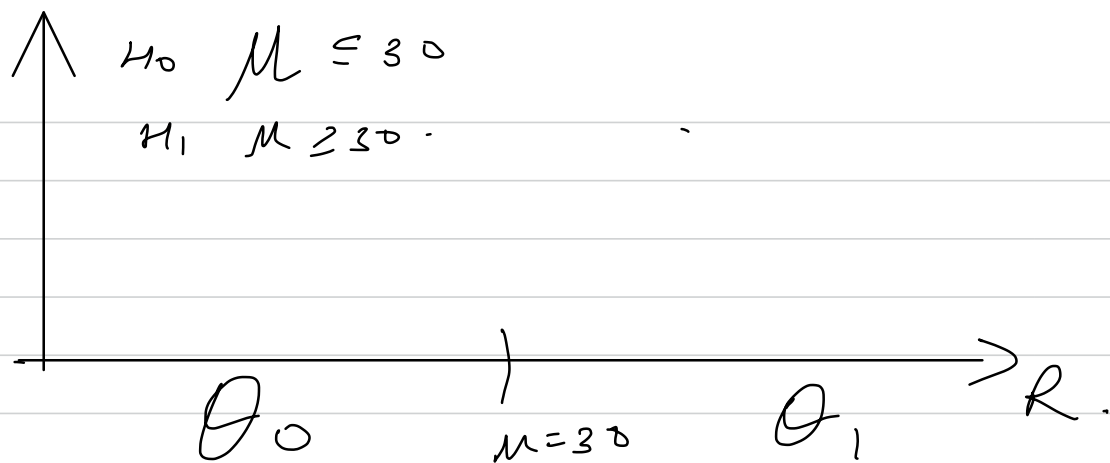
$$\theta \rightarrow P_\theta(\psi=0).$$

↙
1st column $\theta_0 \rightarrow 2^{nd}$ row.

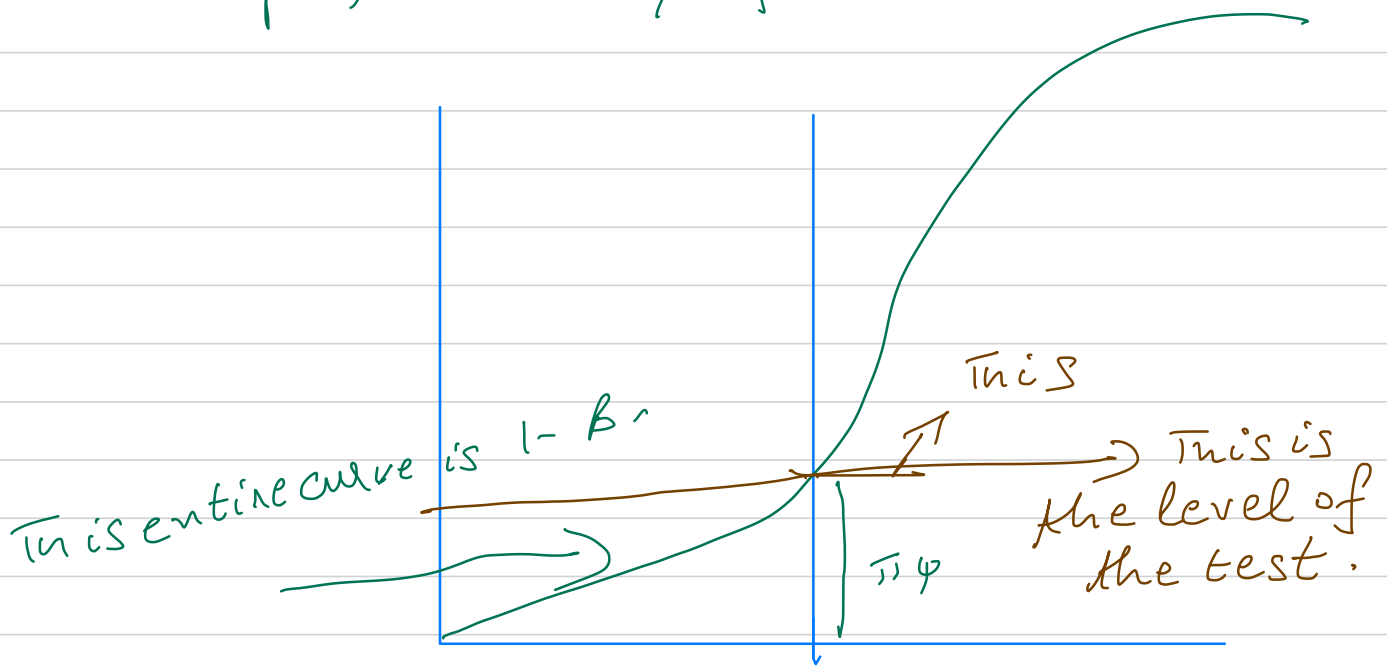
Just a Defn. → Just 1- Type II.
Power of a test:

$$\pi_\psi = \inf_{\theta \in \Theta_1} (1 - \beta_\psi(\theta)).$$

↙
This is just 1-Type II error.
So that's the Probability you reject.
when you really should-



If I start drawing the probability that a set rejects, Power $\pi(\mu)$ of a test.



The Power is the smallest value $1 - \beta$ can take.

6.16. Level of a Statistical Test.

A test φ has level α if

$$\alpha \varphi(\theta) \leq \alpha. \quad \forall \theta \in \theta_0.$$

Think $\alpha = 5\%$, 1% .

You need to control the probability of Type I error.

If I have an innocent person, I want the probability that he be sent to jail to be at most α .

$$\alpha \varphi(\theta) \leq \alpha.$$

H_0 : Innocent

H_1 : Guilty.

I want my test to have
• type I error.

→ Let everyone go.

→ Example.

$$\alpha \geq \alpha \varphi(\theta).$$

How can I make a test with 0 Type 2 error?

A guilty person go free.

Solⁿ:

Send everyone to jail.

Do your best to control that Type I error is at most α .

If that means that you let some guilty people walk free, that's the reality.
 best you can go

So, we are going to make this inequality an equality.

only possible Asymptotically,
Asymptotic level.

$$\lim_{n \rightarrow \infty} \alpha_{\psi, n}(\theta) \leq \alpha \quad \forall \theta \in \Theta.$$

$$\psi = I \int T_n \geq c \, d\gamma.$$

$$\therefore R\psi = \int T_n \geq c \, d\gamma.$$

You have other tests like

$$\psi = \mathbb{I} \{ |T_n| > c \}$$

$$\psi = \mathbb{I} \{ |T_n| \leq c \}$$

It's okay if you tell me the test statistic is an absolute value of $|T_n|$, it's not a universal language, for what is the test statistic

and what is the threshold.



What is really important here is the form of the rejection region.

T_n will involve some re-normalisation of $\bar{x}_n$

$$\text{It's okay if } \tau_n = \sqrt{n} \left(\frac{\bar{x}_n - 0.5}{\sqrt{0.5(1-0.5)}} \right)$$

$$|T| > c. \quad |X|$$

$$|Z| \geq 90.25.$$

$$\lambda = 90.25$$

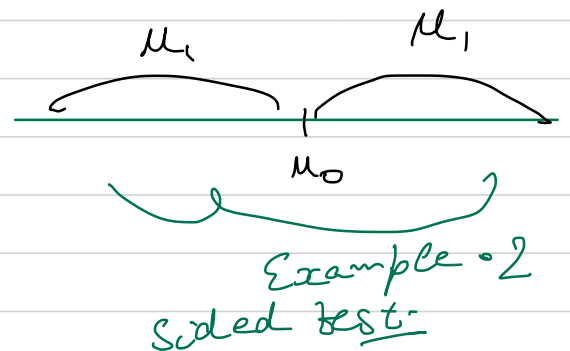
17 One Sided v/s Two sided,

Note: Watch this lecture again.

2-sided:

$$\begin{cases} \mu = 30 \\ \text{v/s } \mu \neq 30 \end{cases}$$

$$\begin{cases} p > 1/2 \\ p < 1/2 \end{cases} \rightarrow \text{2 sided}$$



LECTURE 7:

7.3) 2-sided test of Bernoulli experiment. (example).

→ we will see p-value. → V. Imp

→

These are tests

Test \ Truth		
	H_0	H_1
$\psi = 0$	✓	Type II
$\psi = 1$	Type I θ_0 Probability of Type I: $\alpha(\theta) = P(\psi=1 \theta \in H_0)$	✓

Since we are under $\psi=1$,
if this is not true, it's
certainly not acceptable
and we conclude to H_1 .

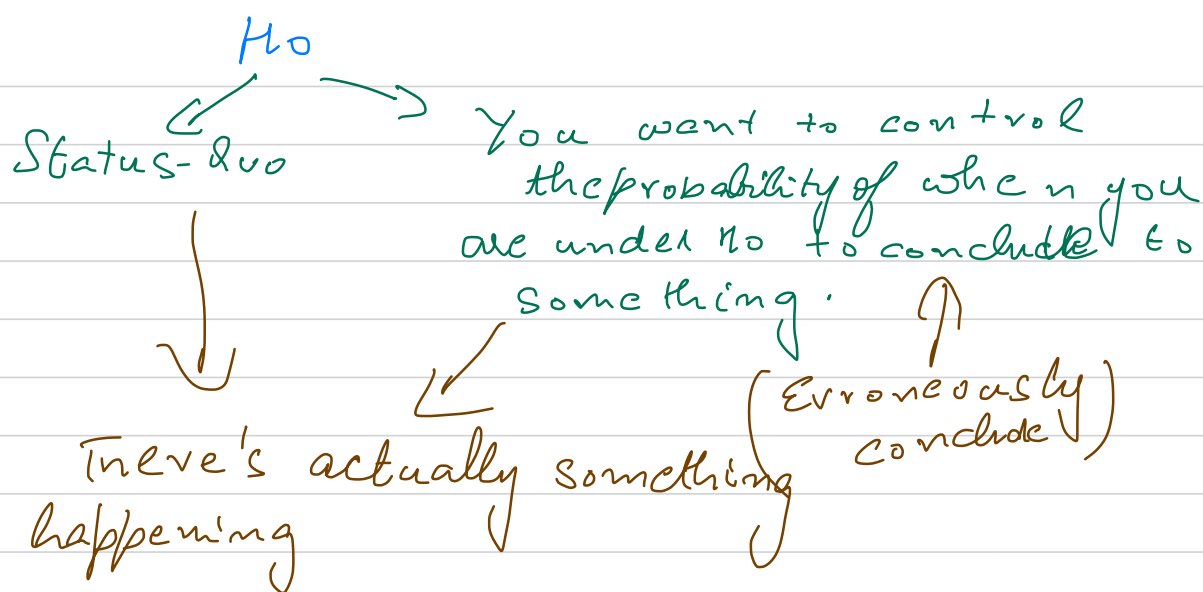
These are Hypothesis.

→ The truth is either H_0 or H_1 . → Reject H_0 , Conclude to H_1 .

→ We never conclude to H_0 , just fail to
conclude to H_1 (reject H_0)

NULL →

ALTERNATE: → Concludes to H_1 (rejects H_0).



$H_1 \rightarrow$

↳ If true we conclude to H_1 , Not a mistake.

$\sup_{\theta \in \Theta_0} \alpha(\theta) \leq \alpha$ → level of the test.
Level α test.

↳ we know for all θ in Θ_0 the max value $\alpha(\theta)$ can take is less than α .

if we are doing asymptotic, we can put the prefix asymptotic.

everywhere we make asymptotic statements.

(probability using CLT as $n \rightarrow \infty$).

ASYMPTOTIC:

Probability of type I:

$$\alpha(\theta) = P_{\theta}(\varphi=1) \quad \theta \in H_0.$$

$\varphi=1 \rightarrow$ means

Ques:- $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Ber}(p) \cdot \quad \forall p \in (0,1) \cdot$

$H_0: p = 1/2$ vs $H_1: p \neq 1/2$.

at level α .

Solⁿ: Let $T_n = \left[\frac{\sqrt{n} (\hat{p}_n - 0.5)}{\sqrt{0.5(1-0.5)}} \right], \hat{p}_n = \bar{X}_n.$

I want my test statistic to be of the form:

$$T_n = \sqrt{n} \frac{\hat{p}_n - p}{\sqrt{p(1-p)}} \xrightarrow[n \rightarrow \infty]{(d)} N(0,1)$$

$$\sqrt{n} \bar{X}_n - p$$

If I say this is my test static, I observe

$p \geq 0.5$ by far, I claim $p \neq 1/2$
otherwise $p = 1/2$.

When we compute a test statistic, we have to control the level; so we compute this probability only under the null and then we do the alternative.

Again, THE ONLY WAY TO CONTROL THE LEVEL, i.e., COMPUTE THE VALUE OF TYPE I ERROR IS NEVER REJECT, NEVER CONCLUDE TO H_1 . But we want to be smart.

I have the test of the form.

$$\alpha(p) \rightarrow f\left(\frac{\sqrt{n}(\hat{p}_n - p)}{\sqrt{p(1-p)}}\right) \in R$$

I want my STATISTICAL TEST
To be of this form.

If this is in R , I want to reject

$$\therefore \alpha(p) = P\left[\frac{\sqrt{n}(\hat{p}_n - p)}{\sqrt{p(1-p)}} \in R\right] = \alpha(p).$$

The probability of reject when in $R \rightarrow 0$.

$$p \in H_0 = \left\{\frac{1}{2}\right\}.$$

All possible values of p are just $\frac{1}{2}$.

Plug it in.

$$\alpha\left(\frac{1}{2}\right) = P\left[\frac{\sqrt{n}(\hat{p}_n - \frac{1}{2})}{\sqrt{\frac{1}{4}}} \in R\right] = \alpha.$$

level
substit

The supremum over all possible values of α
when the range in θ_0 is also $\{1/2\}$.
That's the worst possible value.

NOTE: If $p \in (0,1)$. I want to pick p so that
 $\sup \alpha(p)$ is the lowest for $p \in (0,1)$.

I want

$\sup_p \alpha(p) < \alpha$. $\rightarrow$ Since
I can be smart again.
 $\swarrow$
Never reject.

$\{0,1\}$

$\sup(\alpha(p)) = \alpha$. θ_0 .

$\rightarrow$ I want to pick my Rejection Region s.t.
The possibility of Rejecting $= \alpha$.

$\alpha(\theta)$

$$\psi = 1 \quad \text{if} \quad \sqrt{n}(\hat{p}_n - 1/2) > C$$

↑
Threshold

I want.

$$\begin{aligned} T_n &\rightarrow \alpha. \\ \sqrt{n}(\hat{p}_n - 1/2) &> C \\ \sqrt{n}(\hat{p}_n - 1/2) &> \sqrt{n} \left(\frac{q\alpha/2}{\sqrt{1/2}} \right) \\ &\approx \sqrt{n} \cdot q\alpha/2 \end{aligned}$$

$$T_n \rightarrow \alpha. \\ \sqrt{n} \left(\frac{q\alpha/2}{\sqrt{1/2}} + \frac{1}{2} \right)$$

$$\begin{aligned} & \circ \circ, \quad p_{1/2} \left[\sqrt{n} \frac{|\hat{p}_n - 1/2|}{1/2} > q\alpha/2 \right] \Rightarrow \alpha. \\ & \quad \uparrow \quad \quad \quad \uparrow \quad \quad \quad \uparrow \\ & \quad p=1/2 \quad \quad \quad 1/2 \quad \quad \quad \text{Threshold} \quad \quad \quad \text{level.} \end{aligned}$$

$$\sqrt{n} |\bar{X}_n - 1/2| > q\alpha/2 \quad \text{Test} \rightarrow \alpha.$$

$$\frac{\sqrt{n}(\bar{X}_n - 1/2)}{q\alpha/2 - 1/2} > q\alpha/2$$

So, the test ψ with rejection region.

I can describe
my test, any way I want
simplest $\rightarrow$ Rejection region -

$$Z \geq |x| \\ 1 - \Phi(|x|) \\ 1 - 2.5^{-\Phi(|x|)}$$

$$R_\psi = \{ 25n |\bar{X}_n - 1/2| > q_{\alpha/2} \}$$

has asymptotic level α .

$\rightarrow$ If H_0 is true, then by CLT -

$$P[T_n > q_{\alpha/2}] \xrightarrow{n \rightarrow \infty} 0.05$$

$\alpha = 0.05$

only true.

Under H_0 If I say, the reality has to be the null, and that's the only case in which I am interested in computing this probability

$$Z \geq x \quad \text{and} \quad Z \leq -x \\ 1 - \Phi(x) \quad \text{and} \quad \Phi(-x) \\ 1 - 2\Phi(-x) \\ 1 - 2\Phi(-x) \\ P(|Z| \geq q_{\alpha/2}) = \alpha$$

NOTE:

You could say p is not equal $1/2$.

Because the relationship between mean and variance is somewhat violated.

And by the way, you have this relationship b/w μ and σ^2 by doing this. ok!

So Here, $\theta = 1/2$, that $\psi = 1$, That's the probability I reject.

α_{ψ} is when I reject the NULL.

$$Z < |x|$$

In REAL world:

Fair Coin Real world data;
13 / 30 coin tosses.

$$\rightarrow \sqrt{n} (\bar{X}_n - 0.5) / 2 > z_{\alpha/2} = z.$$

Here =

$-z$.

$$\bar{X}_n = 0.77.$$

$$R_p \left(\sqrt{n} (\bar{X}_n - 0.5) / 2 > z_{\alpha/2} \right) = 0.77 < 1.96$$

$z_{\alpha/2}$ $\alpha = 0.05$

$$z_{\alpha/2} = 1.96.$$

$$\text{when } \alpha = 0.05 \quad z_{\alpha/2} = 1.96.$$

$$\text{when } \alpha = 0.05.$$

$$z_{5\%} = 1.645$$

$$z_{\alpha/2} = 1.96.$$

$$\therefore R_p \left(\sqrt{n} (\bar{X}_n - 0.5) / 2 > 0.77 < 1.96 \right).$$

$\therefore$ we fail to reject.
↳ Status Quo.

If we did not have absolute value.

our cut-off $q_{\alpha/2} = 1.645$, which is for a one sided test.

↓
Eg → news on Youtube;

→

→ You want to reject. Failing to reject gives us no new information and therefore, we always want to reject.

$\alpha/2$ If this was 2.75, α would get even lower.
I would fail to reject this

STUPID STUPID TEST.

$$P\left(\frac{\bar{X}_n - \theta}{1} > q_{\alpha/2}\right) = \alpha,$$

$$\bar{X}_n > \frac{q_{\alpha/2}}{\theta}$$

7.5, ERRORS, LEVEL AND CONCLUSION OF A ONE SIDED TEST;

News on Youtube:

$$\sqrt{n} \frac{\hat{p}_n - p}{\sqrt{p(1-p)}} < c.$$

Here $H_0: p \geq .33$ vs $H_1: p < .33 \rightarrow$ One sided test.

$$P_p \left[\sqrt{n} \frac{\hat{p}_n - p}{\sqrt{p(1-p)}} < c \right] \xrightarrow{n \rightarrow \infty} \alpha(p).$$

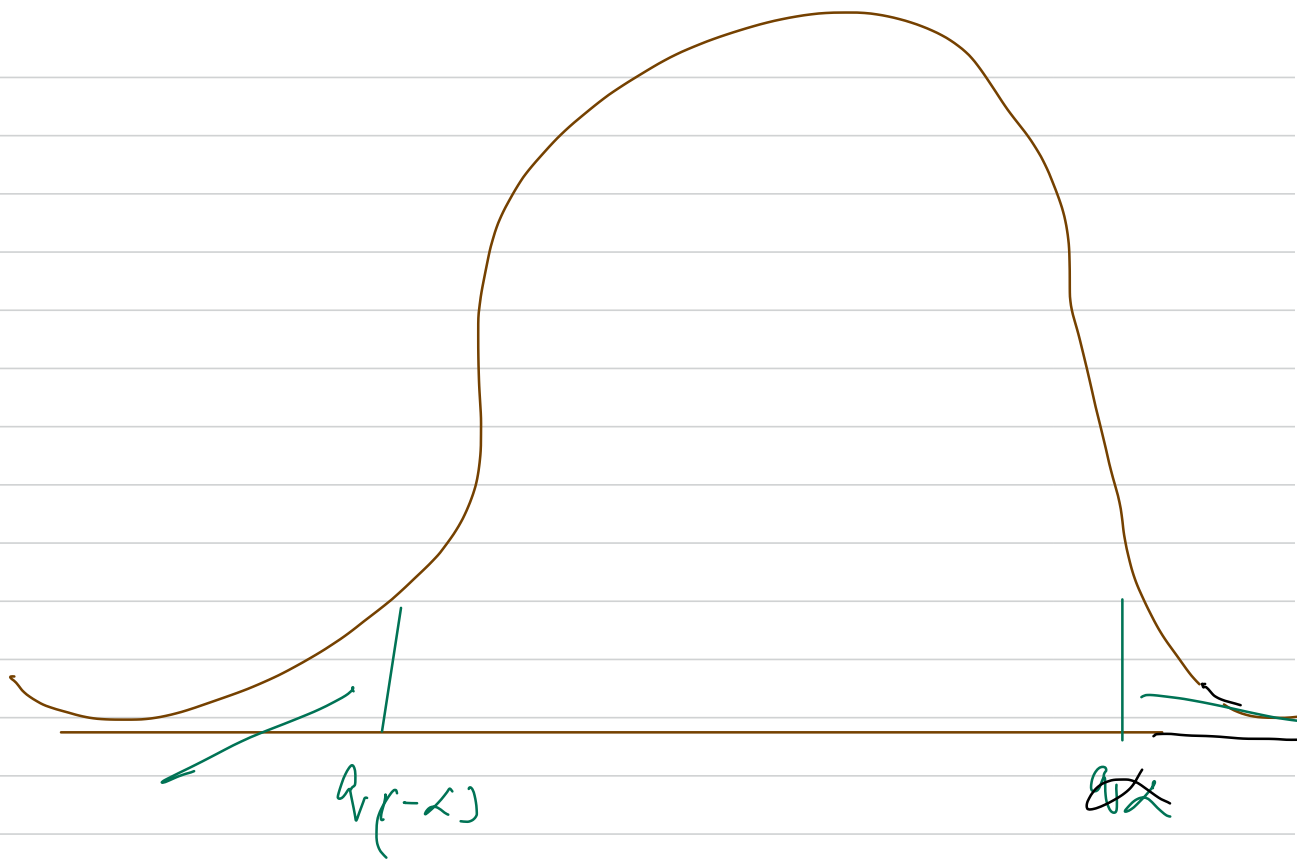
$$\sup_{p \in \theta_0} \alpha(p) = \alpha \rightarrow \text{Level test.}$$

$$p \in \theta_0 = [0, .33, 1]$$

Question what should be the value of c ?

$$\text{Now, } P_p \left[\sqrt{n} \frac{\hat{p}_n - p}{\sqrt{p(1-p)}} < c_{(1-\alpha)} \right] \rightarrow \alpha.$$

If $p = 0.48$ I could plug it in, once the numbers come in and check.



$$q_{1-\alpha} = -q_{\alpha}$$

But this is only true for $p=0.5$.

p

ϕ

$$P(X > q_{\alpha/2}) = \alpha$$

Solⁿ:

this is now a new a no.

$$P_{p_0} \left[\sqrt{n} \frac{\bar{X}_n - p_0}{\sqrt{p_0(1-p_0)}} < -z_{\alpha} \right] = f(p, p_0) \leq \alpha.$$

for all $p \geq .33$

For my notion of one-sided test.

→ I want this probability, so that no matter what p_0 I choose, no matter what p is, asymptotically less than α .

That's the definition of Type I error.

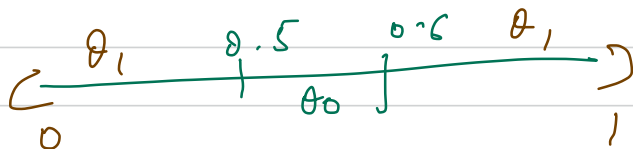
I want to pick those p 's that is true for any p in H_0 .

$$f_n(p_0), \sup_p P_p \left[\sqrt{n} \frac{\bar{X}_n - p_0}{\sqrt{p_0(1-p_0)}} < -z_{\alpha} \right] \xrightarrow{n \rightarrow \infty} \alpha$$

I want take $p_0 = 0.33$ (at the boundary).

$$H_0: p \in [0.5, 0.6]$$

$$H_1: p \notin [0.5, 0.6]$$



Let's start from scratch the way to build a test statistic:

$$H_0: p \leq .33 \quad \text{vs} \quad H_1: p > .33$$

Common-sense: Reject if $\hat{p}_n = \bar{X}_n > \lambda$.

For λ to be chosen later.

$$\sup_{p \leq .33} P_p[\bar{X}_n > \lambda] \xrightarrow{n \rightarrow \infty} \alpha.$$

$$\sup_{p \leq .33} P_p \left[\sqrt{n} \frac{\bar{X}_n - p}{\sqrt{p(1-p)}} > \frac{\sqrt{n}(\lambda - p)}{\sqrt{p(1-p)}} \right] = \alpha.$$

$N(0,1)$.

$C' \rightarrow$ if $I p \in C$ or if I

pick λ it's just one and the same thing 1 degree of freedom.

The p-value of a statistical test:

→ p-value is what people produce in their papers, when they say, I've made a discovery from my data or not, rather than saying "YES OR NO".

→ The numerical computation, is not the value of the test-statistic. It's not a number which is 0 or 1. It's actually a p-value.

"p-value is something which lives on the same scale on which α lives."

↳ How about I did this test at 5% or at 10%.

→ Rather than putting a cut off on the space where my test statistic is

↙ The defⁿ is bit of a mouthful:

The space of Gaussians which is the x -axis of the Gaussienⁿ.

→ It is actually the area under the curve.

p-value:

$$\psi(\alpha=1)$$

done at
test α

The (asymptotic) p-value of a test ψ_α is the smallest (asymptotic) level α at which ψ_α rejects H_0 . It is random, it depends on the sample.

Golden-Rule:

p-value $\leq \alpha$: H_0 is rejected at asymptotic level α .

As α increases, I am more and more likely to reject. As α goes down to 0, I may stop rejecting after a while. If you go above this level, you reject.

When α is 0, I will never reject.
When α is 1, I will reject if I want.

$$P\left[\frac{\sqrt{n}(\hat{p} - p_0)}{\sqrt{p_0(1-p_0)}} < -z_\alpha\right] \leq \alpha.$$

As α goes from 0 to 1, I'm moving z_α from $-\infty$ to ∞ .

There comes a value, when I decide to stop rejecting and move to rejecting.

The p-value is 4%; it means you reject at 5%.

If p-value is .5%, you can reject 1%, and

$$P(Z > |z|) = 1 - P(Z < |z|) \quad \text{above.}$$

If my software says, 10^{-4} , you can think of it as 0.

$$P(Z > |z|) = 1 - \Phi(|z|) = 1 - \Phi(1.96) = 1 - 0.975 = 0.025$$

→ Scientific discoveries associate p-value, to a number that's very small.

~~You would like to think your p-value of as probability of type I error.~~

But you would like to think your p-value small.

If my p-value is 0.01%, I will reject any value above that.

It gives you a number,
Either ~ reject
Fail to Reject.

How strongly did you reject?

7.8 The p-value of a 2 sided test:

Kiss Example:

$$X_1 \dots X_n \stackrel{iid}{\sim} \text{Ber}(p) \quad (p \in (0,1)).$$

$$\left. \begin{array}{l} H_0: p = 1/2 \\ H_1: p \neq 1/2 \end{array} \right] \rightarrow \text{No doubt why}$$

$$\frac{\sqrt{n} |\bar{X}_n - 1/2|}{\sqrt{(1/2)(1/2)}} = \sqrt{n} |\bar{X}_n - 1/2|.$$

Reject at level α if

$$2\sqrt{n} |\bar{X}_n - 1/2| > z_{\alpha/2}$$

$$n = 124$$

$$\hat{p}_n = 0.645 \quad \alpha = 0.05$$

$$= 3.228 > 1.96 \rightarrow \text{Reject}$$



It is in the tail of the
Gaussians.

(But this is only true
for Gaussians)

p-value! what is a p-value?

$$P_{1/2} \left(\frac{\sqrt{n} (\bar{x}_n - \mu_2)}{\sqrt{1/2(1-1/2)}} > 3.229 \right) = \text{p-value}..$$

$\Downarrow$
 $N(0, 1)$

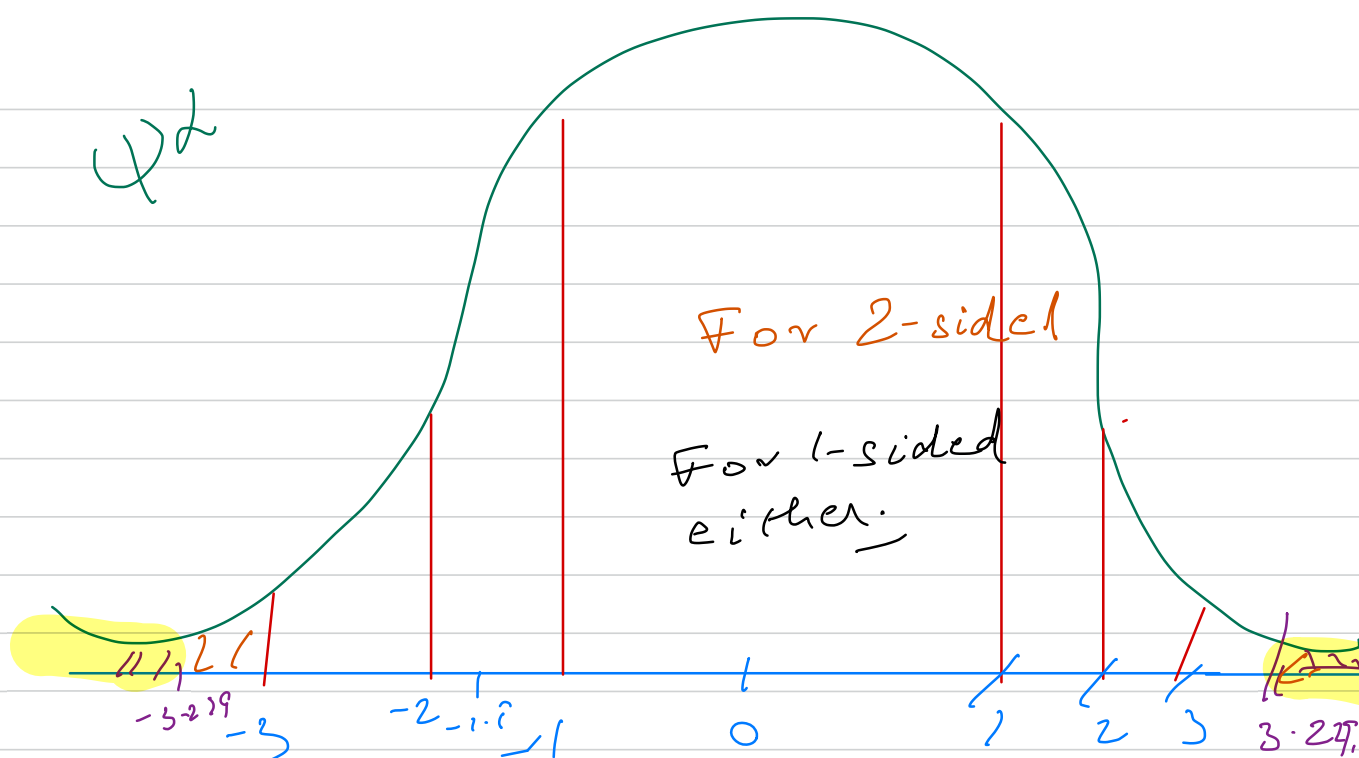
So, p-value = $P(|Z| > 3.229)$ where $Z \sim N(0, 1)$

The way you compute it is, first compute the numerical value your test statistic actually takes.

Then, substitute the C , the threshold with this value.

$$2\phi(x) > \alpha_{1/2} \cdot 2.5$$

$$P(|Z| < \alpha_{1/2}) = \alpha.$$



And the way we say it is just by looking at the picture above, the probability that I am either to the left of it or to the right of it is extremely small.

How small?

That's precisely my p-value.

It would be either this area under the curve or

↓
The yellow shaded area under the curve.

If we compute the area it's going to be less than 10^{-4} , because it's very hard to compute. It's going to be very small.

↓

So small, no-matter what area you come up with, it's going to be less than α .

$$\begin{aligned}
 Z < x & \quad P(Z < |x|) & P(Z < |x|) \\
 & \quad + \Phi + 3 - x & = P(Z < x) \\
 & \quad + 1 - Z < x. & \quad \Phi(|x|) - \alpha_{2,5} \\
 & \quad + 1 + 2\Phi & P(Z > |x|)
 \end{aligned}$$

$$P(|Z| > x) = 1 - (2\Phi(x) - 1) = P(Z > x) + P(Z < -x)$$

$$\begin{aligned}
 P(|Z| < x) & = 1 - P(|Z| > x) \\
 & = 1 - (2\Phi(x) - 1) = 2\Phi(x) - 1 \\
 & = 1 - \Phi(|x|) - \alpha_{2,5}
 \end{aligned}$$

$$\begin{aligned}
 |Z| & \quad P(Z) < x & P(Z) < x \\
 & \quad \leq \alpha & = 1 - \Phi(|x|) - \alpha_{2,5} \\
 & \quad \Phi(|x|) - \alpha_{2,5} & \Phi(PZ < |x|) \\
 P(Z > \alpha_{2,5}) & \quad \Phi & P(Z < x) + P(Z > -x < Z < x) \\
 & \quad \Phi & 2\Phi(x) - 1 - \alpha_{2,5}
 \end{aligned}$$

Exercise: Cookie:

(Refer to the slide & the question)

Stu

Question:

Students are asked to count the number of chocolate chips in 32 cookies for a class activity.

They found that the cookies on average had 14.77 chocolate chips with $\sigma = 4.37$.

chocolate chips. The packaging claims, there are at least 20 chocolate chips per cookie.

One student thinks the number is unreasonably high since the average they found is much lower.

Another student thinks the difference might be due to chance.

What do you think?

Solⁿ:

$$n = 32 \quad \bar{x}_n = 14.77.$$

$$\sigma = 4.37.$$

Here, we are going to make some modelling assumptions.

What should be the right model for this?

$n \geq 30$. We could have thought of Poisson but since
 $\therefore$ We are going to assume Gaussian.

Cookies:

X_i are iid $N(\mu, \sigma^2)$.

I am going to assume σ^2 is known
and it is

$$\sigma^2 = (4.37)^2. \quad \bar{X}_n = 14.77.$$

Now, I want to perform a test
about μ .

$$H_0: \mu \geq 20. \quad \text{vs } H_1: \mu < 20.$$

Status Quo.

$\uparrow$
(this was established)
by printing on the box.

It could be, that nothing was printed
on the box and some students went
to the packaging company and said